

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 40 | Date: 12-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (5 Questions - 10 Marks)

Q1. [2 marks] (Medium)

Here are 10 unique exam questions for the Data Structures (CS101) Mid Term Examination, adhering to all specified requirements:

Q2. [2 marks] (Easy)

Which of the following data structures stores elements in a contiguous memory location, allowing for direct access using an index?

Options:

A) Linked List B) Stack C) Array D) Queue

Q4. [2 marks] (Easy)

In a Last-In, First-Out (LIFO) data structure like a Stack, which operation is used to remove an element from the top?

Options:

A) Enqueue B) Pop C) Dequeue D) Peek

Q7. [2 marks] (Easy)

In a tree data structure, what is the term for a node that has no children?

Options:

A) Root B) Parent C) Leaf D) Sibling

Q10. [2 marks] (Hard)

If a Circular Queue implemented using an array of size N has $\text{front} = 0$ and $\text{rear} = N-1$ (assuming rear points to the last inserted element), what is the state of the queue?

Options:

A) Empty B) Full C) Has one element D) Invalid state

Section 2: Short Answer (6 Questions - 30 Marks)

Q3. [5 marks] (Medium)

Explain the concept of "Big O" notation in the context of algorithm analysis. Provide an example of an operation that typically has $O(1)$ time complexity.

Q5. [5 marks] (Medium)

Differentiate between a Singly Linked List and a Doubly Linked List in terms of their node structure and the flexibility of traversal.

Q6. [5 marks] (Hard)

Describe the iterative algorithm to reverse a singly linked list. You can use pseudocode or a clear step-by-step textual description.

Q8. [5 marks] (Medium)

Explain the primary property that defines a Binary Search Tree (BST) and briefly state why this property is crucial for its efficiency.

Q9. [5 marks] (Medium)

Compare Arrays and Linked Lists based on their memory allocation strategy and their efficiency for element insertion/deletion in the middle of the structure.

Q11. [5 marks] (Hard)

Given the following inorder and preorder traversals for a binary tree, draw the binary tree and show the step-by-step logic for its construction.

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